INFT1004 – Introduction to Programming

University of Newcastle
Course Coordinator: Dr. Shiva Pedram

Assignment 2

Campus Equipment Hire System – Part 2

> **Weighting:** Marked out of 100. See the separate rubric.
> **Submission:** One Python file via the Assignment 2 link on Canvas

Introduction

This assignment builds on the Campus Equipment Hire System you developed in Assignment 1. In that version the stock levels were lost as soon as the program closed, and the desk could only handle the four categories that were written into the code. Here you will fix both of those limitations.

The objective is to extend your program by integrating an external CSV file for data handling, using exceptions for error handling, and using a dictionary to manage the hire data. You may also add a data visualisation using the matplotlib library, which is optional.

Note: the Restock and Hire Summary actions from Assignment 1 are not required in this assignment. The menu has been redesigned as described below.

Carefully read the specification below. Make sure you have all the information necessary to start writing the program. If you are uncertain of something, do not make assumptions. Instead, post your questions to the "assignment2" discussion board forum and check it regularly, or ask your demonstrators during your lab.

Your solution must be your own work. You may be required to participate in a short oral interview of approximately 10 minutes, where you will be asked questions about your code and how it works.

Core Rules

- The program must be menu-driven and continue running until the user chooses Quit.
- All quantities entered by the user must be positive integers. No negatives and no decimals.
- All inputs must be validated so that the program does not crash on invalid input.
- The stock must be loaded from HireInventory.csv at the start of the program. You may use the csv module or read the file as plain text.
- Category names must be handled case-insensitively across all menu actions. A student typing “bikes” should get the same result as “Bikes”.

Data File Requirements

Your program must read from an external CSV file named HireInventory.csv. This file will be provided on Canvas. It has the following structure:

```
Category,Count
Balls,20
Rackets,15
Bikes,10
Tents,18
```

Assume the values inside the CSV are correct, but you must handle cases where the file cannot be read.

File error handling (required)

- You must use try/except to catch FileNotFoundError.
- If the file is not found, print "File was not found" and the program must exit gracefully.

Program Startup

When the program starts:

1. Open HireInventory.csv and read its contents.

2. Store the stock in a dictionary, for example {"Balls": 20, "Rackets": 15, ...}. Values read from a file arrive as strings, so remember to convert them to integers.

3. Store the starting quantities as the maximum for each category in a separate dictionary, for example {"Balls": 20, ...}. These maximums must be preserved for the entire program run and must not be overwritten when the stock is updated or saved.

> **Why keep a separate maximum dictionary?**
>
> Action 4 uses the original starting quantity as the maximum for stock status calculations. If a category is added during the session, the quantity entered at that point becomes its maximum. When the program saves and the user restarts, the CSV will contain updated counts, so the maximums have to be captured at load time and kept separate.

Menu Options

After loading the stock, the program must repeatedly display a menu until the user quits. If the user enters an invalid menu option, display an error message and re-prompt.

Action 1 – Display Current Stock

- Print all categories and the current number of items in each.
- Output should be clearly formatted and easy to follow.

Action 2 – Add New Category

The hire desk sometimes buys gear it has never stocked before. The user must enter the name of the new category, for example "Kayaks", and the number of items purchased.

Validation requirements

- The category name must be unique. It must not already be in the stock list.
- If the category already exists, print an error message and re-prompt until a valid new category is entered.
- The initial number of items must be a positive integer.
- After successfully adding the new category, print the updated list of categories and their stock levels.
- The initial quantity entered becomes the maximum for this category in status calculations, and must be stored in the maximum dictionary.

Action 3 – Hire Out or Return Items

This menu item allows the user to either hire items out or take them back. Note that the user can now process several items at once, for example a sports club collecting ten balls for training. This is a change from Assignment 1, where every action moved a single item.

Step-by-step behaviour

1. Display the current stock.
2. Ask the user whether they want to hire out (H) or return (R). If any other input is entered, display an error message and re-prompt.
3. Ask the user to enter the category name and the quantity, which must be a positive integer.

Validation requirements

- The category must exist in the stock list. Case-insensitive matching is required.
- The quantity must be a positive integer.
- For a hire, the requested quantity must not exceed the available stock. If there is not enough stock, display a message and re-prompt.
- For a return, the stock increases accordingly. Returns are always allowed and no maximum capacity applies in Assignment 2.
- Because returns have no cap, stock can rise above its maximum. In that case the percentage calculated in Action 4 will be over 100%, and the status is High.

After a successful hire or return

- Update the stock.
- Print a confirmation message showing the updated stock for that category.

Exception handling

- Invalid input must not crash the program.
- Invalid menu or action choices must display an error and re-prompt.

Action 4 – Stock Analysis with Optional Visualisation

This option produces the report the centre manager uses each month. It must provide a text report. A visual report is optional.

Text report

Calculate and display:

- The total number of items held across all categories.
- The average number of items per category. If the result is a decimal, round up to the nearest whole number, so 2.5 becomes 3. You may use math.ceil() for this.

You must also display the stock status for each category:

- High: 75% or more of the maximum
- OK: 50% to under 75% of the maximum
- Low: under 50% of the maximum

Percentages must be calculated as (current stock ÷ maximum) × 100. The maximum for each category is the original starting quantity loaded from the CSV, or the initial value entered when a category was added during the session. These values come from your separate maximum dictionary.

Results must be clearly labelled and readable. A loop over the stock dictionary is all that is needed. No external package is required for these calculations.

Visual report (optional)

The bar chart is optional. You will not lose marks if you do not include it. Matplotlib is covered in week 9, so if you would like to add it, create a bar chart showing:

- Categories on the x-axis.
- Current stock counts on the y-axis.

- A chart title and labelled axes.
- The chart displayed on screen.

Action 5 – Save Changes to CSV

- Write the current stock back to HireInventory.csv so that it persists after the program ends.
- Keep the same Category,Count header row and format as the original file.
- Use try/except to prevent crashes during file writing.
- If saving is successful, print a confirmation message.

Action 6 – Quit

- The program must automatically save the latest stock to HireInventory.csv, even if Action 5 was not used. Staff should not lose a day of hires by forgetting to save. Apply the same exception handling used in Action 5.
- Display a farewell message.
- Exit the program safely.

Program Requirements

Required functions

You must implement a separate function for each menu action:

- load_inventory_from_csv()
- display_inventory()
- add_new_category()
- hire_or_return()
- inventory_analysis_and_visualisation()
- save_inventory_to_csv()
- menu()

Required libraries

- **math** — for rounding the average up with math.ceil().

You will also need try/except (week 6) for file handling and input conversion throughout the program. This is part of Python itself rather than a library.

Optional libraries

- **matplotlib.pyplot** — only needed if you include the optional bar chart in Action 4 (week 9).
- **csv** — you may use it to read and write HireInventory.csv, but reading and writing the file as plain text is equally acceptable.

The pandas package is not required and is not assessed in this assignment.

Adapting Assignment 1 code

Your hire, return and analysis logic from Assignment 1 should be carried over and adapted to work with the new dictionary-based data structure and CSV loading approach. You do not need to include Restock or the Hire Summary.

Program Comments

At the beginning of your program, include the following header:

> Author: <your name>
> Date: <start date> – <upload date>
> Task: INFT1004 Assignment 2 – Campus Equipment Hire System – Part 2

Comments are expected at a function level, explaining the purpose of each function, and for any non-obvious inline logic. Comments should be meaningful rather than trivial.

What to Submit

Submit your Python file called <YourName>Assign2.py via the Assignment 2 link on Canvas. For example, if your name is Abby Zammit, your file will be called AbbyZammitAssign2.py.

Marks will be awarded for layout, both visual (variable names, indentation) and structural (scope of variables, use of functions); documentation (comments); and ability of the submission to perform as specified. A detailed marking rubric is available on Canvas.

Late Submission and Adverse Circumstances

Late submissions will not be accepted. Work submitted after the due date will not be marked unless an extension has been approved through the University's adverse circumstances process.

If adverse circumstances affect your ability to submit on time, apply for special consideration as early as possible. See https://www.newcastle.edu.au/current-students/learning/assessments-and-exams/adverse-circumstances for more details.

Questions

In Canvas you will find a forum in the discussion board called “assignment2”. Any question about assignment 2 you can post there. You can also send me an email: shiva.pedram@newcastle.edu.au